Anatomy and Repair of a Native-Language "Erosion": Prompt Optimization and Token Budgets

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Abstract

Automatic prompt optimizers such as DSPY's BOOTSTRAPFEWSHOT are marketed as turnkey accuracy boosters and increasingly applied to low-resource languages. In earlier work we reported that BOOTSTRAPFEWSHOT erodes the native-language subject (ONA_TILI, Uzbek grammar and literature) of a subject-partitioned university-entrance benchmark while aggregate accuracy stays flat (pooled McNemar p=0.011), and that the intuitive mechanism—demonstration subject-skew—is not the cause. This paper identifies what causes it, corrects two of our own claims, and repairs one route to the harm—while showing a second route remains. With a fully instrumented harness (raw completions, token usage, finish reasons) we show: (1) one route to the "erosion" is a *deployment-regime artifact*: correctness-based selection sometimes imports verbose reasoning-subject demonstrations that violate the prompt's own brevity instruction, inflate responses into a fixed decoding budget, and surface as truncation and parse failures concentrated on the subject with the longest inputs. Excluding those failures, our original headline weakens to p=0.073. (2) the budget lever is *causal*: raising `max_tokens` from 256 to 2048 removes truncations monotonically (Cochran–Armitage trend $p<10^{-4}$). (3) at scale the native language is *not* intrinsically fragile: a powered within-subject test (n=3,822 paired items, six models, two test sets) finds no knowledge-side erosion—overturning our earlier single-model full-precision result. (4) the effect's sign flips across serving stacks *and* between 4-bit and 16-bit precision in all three large models, because numerics change which demonstrations the optimizer self-selects. (5) a one-line *constraint-consistent* bootstrapping metric plus rescue parsing recovers 85.6% of the erosion while retaining 108% of the math gains, passing a pre-registered bar. (6) on a second language (TURKISHMMLU, six models) the concealment is starker while the mechanism does not travel: aggregate accuracy rises 5.2 points as the native subject gains nothing, and conditional on an item changing answer, native items change for the worse at 2.35× the odds of non-native ones (95% CI $[1.38, 4.17]$, bootstrapping over items), closely matching the 2.27 we obtain on our original stack—yet the absolute native drop is null (p=1.0) and 35 of 37 Turkish native losses are well-formed, in-budget answers that our repair cannot touch. A second, content-side route to the same harm therefore exists, and we do not explain it. Finally, (7) the differential is *not* a BOOTSTRAPFEWSHOT artifact: MIPROV2, which searches instructions rather than selecting correct demonstrations, reproduces it at the same magnitude on the same stack (1.75 vs. 1.63). It also confirms the mechanism from the opposite side—its *more* verbose demonstrations produce a quarter of the format failures, because the instruction it writes enforces the format they violate, which implies a second repair. We release the instrumented harness, per-item logs for every number, and a frozen replication set.

1 Introduction

Chain-of-thought (CoT) prompting of large language models (LLMs) (Wei et al., 2022) helps mainly on math and symbolic reasoning (Sprague et al., 2025) and can hurt elsewhere (Liu et al., 2025). On top of CoT, automatic prompt optimizers—most prominently DSPY (Khattab et al., 2024) with its BOOTSTRAPFEWSHOT teleprompter, which retains as few-shot demonstrations only training examples the model itself answers correctly—are increasingly applied to low-resource languages, where the subject a model is weakest at is typically the users' own language.

In earlier work we reported a small but statistically significant cost hidden by the aggregate metric. On **DTM** (Hazratov et al., 2026), a subject-partitioned Uzbek university-entrance benchmark built from materials of the national State Test Center (*Davlat Test Markazi*, DTM), BOOTSTRAPFEWSHOT left overall accuracy roughly flat while eroding ONA_TILI (Uzbek grammar and literature) in five of six open LLMs (pooled McNemar p=0.011, mean -4.8), and a controlled experiment ruled out the intuitive mechanism, demonstration subject-skew. We closed by conceding the true mechanism was unknown.

This paper answers that open question—and in doing so corrects our own record. We re-instrumented the entire pipeline to log raw completions, token usage, and finish reasons for every generation, then subjected the finding to a causal budget manipulation, a demonstration-length manipulation, a powered within-subject test with a frozen replication set, a fixes shoot-out with a pre-registered success bar, and full-precision (16-bit brain floating point, bf16) coverage of the large models. Our contributions:

- **A mechanism** (§4–§5). Correctness-selection imports verbose reasoning-subject demonstrations that contradict the prompt's own brevity instruction, and under a fixed budget the inflated responses truncate and fail to parse on the subject with the longest inputs. Roughly a third of our original pooled flips were such failures (clean-item p=0.073). The lever is causal: truncations fall 28→16→3→0 across budgets 256→2048 (trend p<10^{-4}), against a flat zero-truncation control.
- **Two corrections** (§4, §7). The observational effect does not replicate across serving stacks (p=0.227), because different numerics select different demonstrations. And our earlier "robustness" result (-7.2, p=0.045, n=250, one model) is overturned by a powered version (n=3,822 paired items, six models): no knowledge-side erosion.
- **Negative controls** (§6). Demonstration *subject* is null, and *length* is null in the mild-payload regime—the harm needs the extreme-payload × tight-budget corner. Models rarely write long native-language reasoning at all, so the budget-breaking register is an import from reasoning subjects.
- **A repair** (§8). Requiring bootstrapped demonstrations to satisfy the prompt's own brevity constraint, plus rescue parsing, recovers 85.6% of the erosion while retaining 108% of the math gains, passing a pre-registered bar.
- **Instability** (§9). Every large model's delta flips sign between Q4 and bf16, and our own tokenizer-fertility hypothesis is falsified by the logs: math *outputs* cost more tokens than Uzbek prose.
- **Generalization, with an honest failure** (§10). On TURKISHMMLU aggregate accuracy rises 5.2 points while the native subject gains nothing, and native items change for the worse at 2.35× the odds of non-native ones ([1.38, 4.17]). But the absolute drop is null and our mechanism does *not* travel: 35 of 37 native losses are well-formed in-budget answers. The harm generalizes. Our explanation of it does not.
- **Not one optimizer's artifact** (§11). MIPROV2 reproduces the differential at the same magnitude (1.75 vs. 1.63) and confirms the mechanism from the opposite side: its *more* verbose demonstrations produce a quarter of the format failures, because the instruction it writes enforces the format they violate.

The upshot inverts our earlier caution without weakening it: the native language is not intrinsically harmed by prompt optimization, but the deployment stack can manufacture a harm that looks exactly like it—and, as Turkish shows, at least one further route to the same harm remains unexplained. Audits must therefore run per subject *on the deployment stack*, score parse failures separately from knowledge errors, and constrain optimizers to respect their own prompts. We release the harness, per-item logs behind every number, and the replication set.

2 Related Work

When CoT helps and hurts. CoT's gains concentrate on math and symbolic tasks (Sprague et al., 2025). It can reduce accuracy on tasks where deliberation hurts humans (Liu et al., 2025), and its marginal value on strong models is shrinking (Meincke et al., 2025). We study the *optimizer* layered on CoT and find its native-language cost flows through format compliance, not knowledge.

Automatic prompt optimization. DSPY (Khattab et al., 2024) and MIPROV2 (Opsahl-Ong et al., 2024) self-bootstrap demonstrations. Sarmah et al.

(2024) conjectured teleprompters over-rely on examples the model already answers well. We measured that skew previously and here show the harm it correlates with is transmitted by demonstration *payload meeting budget*, not subject.

Format restrictions and budgets. Format constraints degrade reasoning performance (Tam et al., 2024). Tokenizers price languages unequally (Petrov et al., 2023), with accuracy consequences across low-resource languages (Lundin et al., 2025). We connect this literature to prompt optimization: the optimizer is a mechanism that *generates* format pressure, and fixed decoding budgets convert it into subject-selective failures—though, contrary to our own initial hypothesis, through input length and induced verbosity rather than output-side fertility (§9).

Cross-lingual reasoning dynamics. A parallel literature asks which language multilingual models reason *in*: latent-language analyses of English-centric internal processing (Wendler et al., 2024; Schut et al., 2025; Zhong et al., 2024), how language-centric pretraining shapes reasoning (Park et al., 2025), and reasoning transfer into extremely low-resource languages (Tran et al., 2025). We do not test these mechanisms, and flag them because they are the most plausible home for the content-side route our own account cannot explain (§10): if correctness-selected demonstrations pull generation toward a dominant-language register, native-language answers could degrade on their merits with no format failure at all. Deciding that needs representation-level evidence we do not collect.

Disparate impact. Zhu et al. (2022) show semi-supervised learning can help high-baseline subgroups at low-baseline subgroups' expense, masked by averages. Our powered test finds the low-resource analogue is, at scale, an artifact of the serving regime rather than a learning-dynamics inevitability.

Uzbek and Turkic evaluation. Native Turkic benchmarks include TUMLU (Isbarov et al., 2025), KazMMLU (Togmanov et al., 2025), and TurkishMMLU (Yüksel et al., 2024). DTM (Hazratov et al., 2026) provides the subject partition our analyses require.

3 Setup

Data. DTM (Hazratov et al., 2026): 1,000 four-option questions (ONA_TILI 400, TARIX 300, mathematics 150, physics 150). Seeded stratified split (train 500/dev 249/test 251, ∼100 ONA_TILI test items per model). During this study we found eight rows of the release file with nulled answers. Two fall in the test split and their gold labels were recovered from our published run logs, six are excluded from train/dev. A verification gate asserts our loader reproduces the original runs' test sequence and gold labels exactly. **Replication set:** the DTM 2019 public corpus contributes 393 deduplicated ONA_TILI items never used in any development decision. Its protocol was frozen before first model contact.

Models and engines. Six open instruction-tuned LLMs spanning GEMMA-4 (E4B, 31B) and QWEN-3.5/3.6 (4–27B). The original study served the models with Ollama at 4-bit quantization (Q4_K_M, hereafter Q4) on an Apple-silicon workstation. The present study replicates that configuration on an NVIDIA DGX Spark and adds vLLM at bf16 for the three large models. Greedy decoding, `max_tokens` 512 unless manipulated.

Conditions. **direct** (answer only), **cot** (brevity-constrained zero-shot CoT: "at most two sentences"), **bootstrap** (BOOTSTRAPFEWSHOT on the CoT module, 4 demonstrations, correctness metric), and **bootstrap-compliant** (the fix: the metric additionally requires the demonstration's reasoning to satisfy the brevity instruction, see §8).

Instrumentation and scoring. Every generation logs raw completion text, prompt/completion token counts, finish reason, and a parse-failure flag. A rescue parser recovers answer letters from unparseable raw text. We score two ways: *deployment* (a parse failure is a wrong answer—what a user experiences) and *knowledge* (items where either condition truncated or failed to parse are excluded). DSPy retries a failed parse through a second adapter before erroring, so all failure counts are lower bounds on incidents.

Statistics. Paired exact binomial McNemar tests throughout, with Holm correction within each preregistered contrast family and Cochran–Armitage for dose trends. All numbers regenerate from committed per-item logs.

Table 1: Per-model ONA_TILI change Δ (CoT$\rightarrow$BOOTSTRAPFEWSHOT) on the original stack (Apple silicon, Q4) and the replication stack (DGX Spark, Q4): same models, seed, and verified split, deployment-scored. The observational effect does not transfer across stacks.

Model	orig. stack	repl. stack
qwen3.5:9b	−9	−5
gemma4:e4b	−8	−3
qwen3.5:4b	−5	−5
qwen3.6:27b	−4	−4
gemma4:31b	−3	+1
qwen3.5:27b	0	+2
pooled McNemar	p=0.011	p=0.227
clean items	p=0.073	p=0.501

4 The Observation, and Its Fragility

Our original headline was a pooled paired erosion of ONA_TILI under BOOTSTRAPFEWSHOT: 75 CoT-correct answers flipped wrong against 46 the other way (McNemar p=0.011, mean −4.8—see Table 1, left column). Instrumenting the shipped traces decomposes those flips: **nine of the 29 net flips were parse failures**—generations the DSPy adapter could not parse, scored as wrong—and they concentrate exactly in the two models with the largest erosions (parse errors: qwen3.5:9b 1$\rightarrow$11, gemma4:e4b 0$\rightarrow$10, on n=100 each). Excluding items where either condition failed, the pooled test weakens to p=0.073. The original claim survives as a *deployment-scored* harm (users did receive unusable outputs) but not, at conventional thresholds, as a knowledge-scored one.

The comparative claim survives item clustering. Two objections apply to the pooled test above. It treats model×item pairs as independent although the same items recur across all six models, and it measures an *absolute* fall in ONA_TILI rather than the comparative claim we actually make, that the native subject is harmed while the others gain. Estimating that contrast directly—the odds that a *changed* item changed for the worse, ONA_TILI relative to the other three subjects, pooled by Mantel–Haenszel with each model as its own stratum and bounded by a bootstrap resampling items as whole clusters—gives an odds ratio of 2.27 on the original stack (95% CI $[1.37, 3.91]$, five of six models agreeing). The interval excludes 1. The comparative form of our original finding therefore holds under the item clustering its critics would apply, even as the absolute form weakens once format failures are removed. §10 turns the same estimator on a second language.

Table 2: Replication-stack means over six models: the prompting ladder redistributes accuracy. On DTM the compliant-bootstrap fix (§8) beats vanilla BOOTSTRAPFEWSHOT on *both* subjects, which §10 shows does not hold on TURKISHMMLU.

Condition	ONA_TILI	math
direct (no prompting)	32.8	53.1
CoT	33.3	74.6
BOOTSTRAPFEWSHOT (vanilla)	31.0	78.5
BOOTSTRAPFEWSHOT (compliant)	**33.0**	**80.3**

The effect does not transfer across stacks. We re-ran the identical protocol—same models, quantization, seed, and a loader verified to reproduce the original test split item-for-item—on different hardware (Table 1, right column). The pooled effect is directionally negative but far from significant (b=65, c=51, p=0.227, clean-item p=0.501), and the two largest original erosions shrink by roughly half. The reason is visible in the logs: BOOTSTRAPFEWSHOT selects whichever training traces the model answers correctly, and numerics differ across stacks, so the *selected demonstrations differ*. On the replication stack, gemma4:e4b's selected payload grew to 3,565 characters (original: 3,080) and its vanilla-bootstrap run produced 62 truncated and 56 unparseable generations across subjects (CoT: 17/13), while models whose selected payloads stayed under ∼1.7k characters produced essentially none. The observational effect, at n≈100 per model, is dominated by which demonstrations the stack happens to select.

The redistribution framing survives. What is stable across both stacks is the *shape* of the effect (Table 2): each rung of the prompting ladder buys large reasoning gains (math +21 from CoT, +4 more from the optimizer) while the native language is flat to slightly negative—and on the original stack, five of six models ended *below the no-prompting baseline* on ONA_TILI after the full stack (means 32.7$\rightarrow$29.3) even as math gained +34. Aggregate accuracy cannot see this. Figure 1 decomposes the replication-stack flips.

5 The Budget Lever Is Causal

If the format component is truncation under a fixed budget, raising the budget must remove it monotonically. It does (Table 3). On gemma4:e4b,

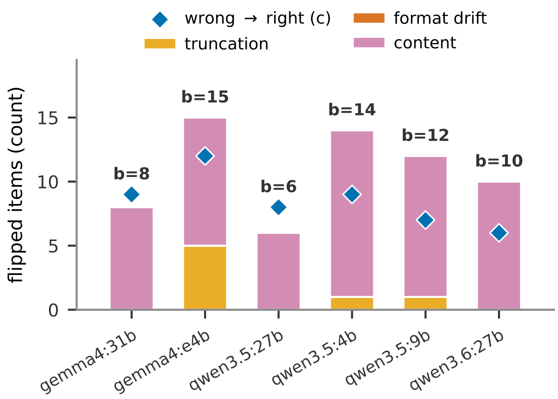

Figure 1: Replication-stack decomposition of CoT-correct→bootstrap-wrong ONA_TILI flips per model into truncation, format-drift, and content classes (markers: opposite flips). Format failures concentrate where selected demonstration payloads are largest.

Table 3: Budget dose-response (ONA_TILI Δ CoT→BOOTSTRAPFEWSHOT, with bootstrap-condition truncation counts on ONA_TILI items). Truncation is causal in the budget. A budget-insensitive residual remains.

`max_tokens`	256	512	1024	2048
gemma4:e4b Δ	−10	−3	−7	−3
truncations	28	16	3	0
qwen3.5:9b Δ	−9	−5	−5	−5
truncations	13	6	3	5
gemma4:31b Δ	0	+1	0	−2
truncations	0	0	0	0

bootstrap-condition truncations fall 28→16→3→0 as `max_tokens` rises 256→512→1024→2048 (Cochran–Armitage trend $p<10^{-4}$) and the ONA_TILI erosion shrinks from −10 to −3. qwen3.5:9b shows the same dose-response (trend p=0.016). The zero-truncation control (gemma4:31b, short selected demos) is flat at every budget. Two readings follow. First, tight budgets are the deployment-worst case: at 256 tokens the erosion nearly doubles. Second, budget relief does *not* eliminate the erosion in the affected models—a budget-insensitive residual of −3 to −5 remains at 2048 with zero truncations, isolating a content-side component that §6 and §7 pursue.

6 Demonstrations: Subject Is Null, Mild Length Is Null

Our earlier controlled experiment varied demonstration *subject* mix and found no effect on ONA_TILI. We previously read this as "the mechanism is more subtle." The instrumented reading is sharper: all demo conditions in that experiment used the model's own correct CoT traces, whose *style and length* are roughly constant across subject mixes—the design inadvertently held the true variable fixed. We therefore ran the missing cell: a 2×2 of demonstration {reasoning, native}×{short, long} (short = satisfying the prompt's brevity instruction), four demonstrations each, four models, paired per-item logs (Table 4).

Table 4: Demonstration composition (2×2 + CoT reference, pooled over four models, ONA_TILI, paired exact McNemar, Holm-corrected). No contrast is significant. The most consistent depression is reasoning-subject demos of *either* length.

Contrast	b	c	p	Holm
reason-short vs. CoT	51	33	.063	.378
reason-long vs. CoT	47	34	.182	.770
native-short vs. CoT	38	33	.635	1.0
native-long vs. CoT	42	29	.154	.770
long vs. short	82	79	.875	1.0
native vs. reason	78	91	.356	1.0

Three results. (1) The *subject* null replicates under paired tests. (2) Demonstration *length* is also null in this regime (p=.875): the mild payloads that models' own compliant-vs-noncompliant traces span (≤1.2k characters) do not reproduce the harm, which lives in the extreme-payload × tight-budget corner that vanilla BOOTSTRAPFEWSHOT occasionally selects (§4). (3) A registry asymmetry emerged that is mechanistically telling: two of four models had *zero or one* long native-subject traces to draw from (versus 6–42 long reasoning traces)—models rarely produce verbose native-language reasoning at all. The budget-breaking register is an import from reasoning subjects, which is why correctness-selection (which over-samples reasoning subjects 2.3×, our earlier Table 2) is the vehicle even though subject labels themselves carry no causal weight. The only directionally consistent cell, reasoning-subject demonstrations depressing ONA_TILI in 4/4 models (p=.063 uncorrected), we report as suggestive.

7 No Within-Subject Erosion at Scale

Is there any knowledge-side erosion once format is out of the picture? Our earlier paper answered yes with its "robustness" result: ONA_TILI-only bootstrapping under vLLM/bf16 eroded the flagship by −7.2 (p=0.045, n=250, one model). We powered that protocol properly: demonstrations bootstrapped from ONA_TILI train only, evaluated

Table 5: Powered within-subject test (pooled over six models, exact McNemar). b: CoT-right→bootstrap-wrong. c: the reverse. No erosion. The benchmark half improves nominally.

Set	Scoring	b	c	n	p
benchmark	deployment	137	171	1464	.060
	knowledge	129	166	1410	.036
replication	deployment	264	260	2358	.896
	knowledge	250	249	2259	1.0

on (i) the benchmark's held-out ONA_TILI items and (ii) the frozen 393-item replication set, across six models and n=3,822 paired items (Table 5).

The verdict is unambiguous: **no knowledge-side native-language erosion under within-subject optimization**. The benchmark half shows a small *improvement*, positive in all six models (+0.8 to +4.9), nominally significant (p=0.036 uncorrected, Holm across the two test sets 0.072). The larger replication set is exactly null. Style analyses of the surviving flips show no assertiveness signature (flip reasonings match stable items in length, hedging, and named-entity density). Our earlier −7.2 should therefore be read as a small-sample fluctuation, and we retract the inference we drew from it. Combined with §4–§6: the harm requires the *cross-subject* regime, where the optimizer can import the verbose register, and a budget for it to collide with.

8 The Repair

BootstrapFewShot's metric checks only correctness, so it happily selects demonstrations whose reasoning violates the prompt's own "at most two sentences" instruction (selected payloads reached 997 characters per demonstration). The fix is one line: *require the demonstration to satisfy the constraint it will teach*. We compare this **compliant** metric, **rescue parsing** (extracting the answer letter from unparseable raw text), their combination, and a 4× **budget** increase, against a bar fixed before running: recover ≥80% of the deployment-scored erosion while retaining ≥90% of the math gain (Table 6).

The combined recipe passes. It collapses the worst model's failure count from 62 truncated/56 unparseable to 22/20 while *raising* its ONA_TILI above CoT (25.0→29.0 vs. CoT 28.0), overshoots CoT on qwen3.5:9b (37.0 vs. 36.0), and retains more than the full math gain on average—because vanilla's format failures were hurting math too (Table 2). Retention above 100% and the trivial 100%

Table 6: Fixes shoot-out (means over six models, recovery of the CoT→vanilla ONA_TILI erosion and retention of the vanilla math gain). Pre-registered bar: ≥80/≥90.

Fix	Recov. %	Reten. %	Bar
compliant metric	75.6	108.3	miss
rescue parsing	57.8	116.7	miss
budget ×4	77.2	180.6	miss
compliant + rescue	**85.6**	**108.3**	**pass**

Table 7: Large models, ONA_TILI Δ (CoT→BootstrapFewShot) at Q4 (Ollama) vs. bf16 (vLLM), same protocol and split. The effect exists at full precision and flips sign per model across precisions.

Model	Q4	bf16
qwen3.5:27b	+2	−5
qwen3.6:27b	−4	+7
gemma4:31b	+1	−5

recovery assigned to models with no erosion are properties of the metric's definition. Per-model values and small-denominator caveats accompany the released tables. The budget fix recovers less, costs 4× the decoding budget, and is better viewed as a complement. The residual it cannot touch—qwen3.6:27b's content-flip −4—matches §6's suggestive reasoning-demo depression, and within-subject bootstrapping avoids it entirely (§7). One caveat travels badly and we record it here rather than in a footnote: on TurkishMMLU the same recipe *costs* reasoning-subject accuracy in five of six models (§10). It remains the only intervention we tested that removes format failures outright, but the retention figure above is a DTM result, not a guarantee. §11 also uncovers a *second* repair we did not design: because the harm is a contradiction between verbose demonstrations and a brevity instruction, strengthening the instruction until it wins removes the tax just as constraining the demonstrations does—and needs no metric change at all.

9 Precision: Not a Q4 Artifact, and a Self-Correction

Our original study ran at 4-bit. A fair question is whether everything above is a quantization artifact. Serving the three large models at bf16 (Table 7): the erosion *appears at full precision* in two of three models (with a truncation component on the flagship), so it is not a Q4 artifact—but every model's delta flips sign between precisions, for the same reason effects flip between stacks: precision changes the correct-set, hence the selected demonstrations. Observational cells at n=100 are hostage to demonstration selection. Audits must run on the deployment configuration, precision included.

Fertility, corrected. We initially hypothesized that Uzbek prose is token-expensive relative to math notation, making fixed budgets effectively smaller for ONA_TILI (cf. Petrov et al., 2023; Lundin et al., 2025). Our logged token counts falsify this for output text: math reasoning costs *more* tokens than ONA_TILI reasoning both per character (∼0.43 vs. ∼0.38) and per item (∼130–160 vs. ∼93–124). The native-language subject is budget-fragile for two other reasons: its *inputs* are the longest (mean 337 characters vs. 156 for math), and imported verbose demonstrations inflate its responses most where models are weakest. We report this correction because the convenient hypothesis was ours.

10 Does It Travel? Turkish

An account this specific should be testable in a second language. We ran the identical protocol—same optimizer, seed, decoding budget, serving stack, and four-subject structure—on TURKISHMMLU (Yüksel et al., 2024), taking Turkish Language and Literature as the native subject against Mathematics, Physics, and History (65 test items per subject, six models, 4,680 generations). Precision is held at Q4 deliberately: §9 shows precision alone flips these models' signs, so varying language and precision together would be uninterpretable.

The aggregate hides more, not less. BOOTSTRAPFEWSHOT raises overall TURKISHMMLU accuracy by 5.2 points (66.4→71.6) while the native-language subject gains nothing (58.5→58.2). Mathematics gains +9.7, Physics +7.2, and History +4.1. Turkish Language and Literature is the only one of the four subjects that does not benefit. A practitioner reading the benchmark score sees a clean five-point win with no indication that the users' own language was left out of it. This is the concealment of §4 in a starker form: on DTM the aggregate stayed flat (−0.1) while the native subject fell, whereas here the aggregate rises while the native subject stands still.

The absolute drop, however, does not travel. Measured as our original study measured it—as a fall in native accuracy—there is no Turkish erosion. The pooled native-subject effect is exactly null

Table 8: Differential harm: the odds that a *changed* item changed for the worse, native subject relative to the others. Mantel–Haenszel odds ratio stratified by model. 95% CI from a bootstrap resampling *items* as whole clusters. Six models each.

Dataset	OR	95% CI	agree
DTM, original stack	2.27	[1.37, 3.91]	5/6
DTM, replication stack	1.63	[0.97, 2.79]	4/6
TURKISHMMLU	2.35	[1.38, 4.17]	5/6

(b=37, c=36, n=390, p=1.0, knowledge-scored b=35, c=36, p=1.0), and no model survives Holm correction. The harm on Turkish is an opportunity cost rather than a loss. Nothing was taken from the native subject, but everything else was added to, and only a subject-partitioned audit can see the difference.

Neither does the mechanism. 35 of the 37 native losses are clean content flips—valid parse, `stop` finish reason, completions far inside the budget. gemma4:31b produces zero truncations in every condition. qwen3.5:27b loses 7.7 points on the native subject with *flat* truncation counts (1/1/1) and takes 0% recovery from our fix. Across models the erosion-vs-failure rank correlation is ρ=−0.27 (exact permutation p=0.62). The mechanism's *scope condition* is nonetheless confirmed: qwen3.5:4b selected the largest demonstration payload of any model on either benchmark (4,387 characters), produced the most failures (17 truncated, 15 unparseable), eroded most (−15.3), and is the only model the fix helps (failures 17/15→5/5, 30.1% recovery). The format tax is real and correctly bounded—on Turkish only one model in six reaches the corner where it fires, and the remaining harm is content-side and unexplained by us.

What travels is the differential. Our claim was never that native accuracy falls in absolute terms—it was that the native subject is harmed *relative to* subjects that gain. That contrast is directly estimable: conditional on an item changing correctness between CoT and vanilla BOOTSTRAPFEWSHOT, are the odds the change is harmful higher for native than for non-native items? Pooling with a Mantel–Haenszel odds ratio (each model its own stratum) and bounding it by resampling items as whole clusters (Table 8), the differential is significant in Turkish and on the original stack, and attenuates on the replication stack—the same stack on which the observational effect attenuates (§4). Computed from the released traces, the original-stack estimate reproduces the exact discordant counts behind our published headline (b=75, c=46). qwen3.6:27b shows why an absolute test cannot see this: its native accuracy *rose* (60.0→63.1) while its odds ratio is 2.4, because its other subjects rose far more. In both benchmarks the native subject is the only one that fails to benefit (Turkish means: History $+4.1$, Mathematics $+9.7$, Physics $+7.2$, native -0.3).

This contrast was **not pre-registered**. The prespecified endpoint is the absolute native-subject McNemar reported above, and we report the differential as secondary. No per-model claim is safe either: qwen3.5:27b reverses between benchmarks (DTM 0.75, Turkish 4.00), and the Turkish estimate is driven substantially by qwen3.5:4b (13.87). Only the pooled, stratified estimate is stable.

The repair is not free here. On DTM the compliant metric retained 108% of the math gains (§8). On Turkish it *costs* mean reasoning-subject accuracy in five of six models (qwen3.5:4b 68.5→61.5, qwen3.6:27b 84.6o80.8). It remains the only intervention we tested that removes format failures outright, and we still recommend it—but it is not the free lunch DTM suggested, and deployments should measure both subjects after applying it.

11 Is It the Optimizer?

BOOTSTRAPFEWSHOT keeps whichever training examples the model answers correctly and never edits the instruction, so the differential could be a property of correctness-based *selection* rather than of prompt optimization. We re-ran the identical protocol (§3)—same split, seed, stack, budget, metric, and 251 items—with MIPROv2 (Opsahl-Ong et al., 2024), which instead searches *instructions* by Bayesian optimization against a validation set drawn from dev.

The differential reproduces. On the same replication stack BOOTSTRAPFEWSHOT gives an odds ratio of 1.63 (95% CI $[0.97, 2.79]$) and MIPROv2 gives 1.75 ($[0.99, 3.14]$, three of four treated models agreeing). Two optimizers sharing almost no machinery land in the same place, so the harm is not an artifact of correctness-based demonstration selection. Suggestively at this n, the one treated model MIPROv2 optimized with *zero* demonstrations is also the only one whose odds ratio falls below 1.

Two models were never treated. Under our search budget (5 candidates, 10 trials) MIPROv2 returned the baseline unchanged for gemma4:31b and qwen3.5:27b—original instruction, zero demonstrations. They cannot speak to the question and are excluded rather than scored as nulls, since the correct statement about them is that our search was too weak to test them, not that the optimizer is harmless there. Effective n is four models.

The mechanism, confirmed from the other side. On gemma4:e4b—our worst format-tax case—MIPROv2's demonstrations are *more* verbose than BOOTSTRAPFEWSHOT's (mean 755 vs. 631 reasoning characters), yet produce 15 truncated and 14 unparseable generations against 62 and 56, and ONA_TILI *rises* 3 points. Payload length alone therefore cannot explain the tax. The difference is the instruction MIPROv2 wrote, demanding "a mandatory, very brief justification" concluding with "*only* the correct single-letter choice"—it enforces exactly the format its own demonstrations violate. Our account says the harm comes from a compiled prompt *contradicting its own brevity instruction*. §8 removes that contradiction from the demonstration side, MIPROv2 removes it from the instruction side, and the tax fails to materialize either way. That is independent support from an optimizer not designed to test it, and it implies a second repair: **strengthen the instruction** rather than constrain demonstrations, with no metric change at all.

12 Discussion and Recommendations

The picture that survives instrumentation, causal manipulation, and power is a **two-regime** account. In the default cross-subject regime, BOOTSTRAPFEWSHOT is an unwitting generator of format pressure: correctness-selection imports the verbose reasoning-subject register, the compiled prompt then contradicts its own brevity instruction, and a fixed decoding budget converts the contradiction into truncations and parse failures landing on the subject with the longest inputs—the users' own language. The size of that harm is set by which demonstrations a given stack's numerics happen to select, which is why it appears, shrinks, and flips across machines and precisions. In the within-subject regime the native language is not eroded at all, at n=3,822.

A second route we do not explain. Turkish (§10) forces a qualification. The differential replicates there—the native subject is again the only one that fails to benefit—but almost none of it runs through format failures: 35 of 37 native losses are well-formed, in-budget, correctly parsed answers that are simply wrong, and our repair recovers none of them. Something about importing correctness-selected reasoning demonstrations degrades native-language answers on their merits, and we cannot say what—though §2's cross-lingual reasoning literature is where we would look first. This is the most important open question we leave: our mechanism is sufficient to produce the harm and sufficient to explain the DTM case, but it is not necessary for the harm to appear.

Four recommendations follow. (1) **Audit per subject on the deployment stack**, at its budget and precision. (2) **Score parse failures separately** from knowledge errors. (3) **Make optimizers respect their own prompts**, by constraining demonstrations or strengthening the instruction (§11). (4) **Prefer within-subject bootstrapping**.

Limitations

Two benchmarks, two languages, and two optimizers, but inference-time prompting only, and reflective or reinforcement-style optimizers may still behave differently. The MIPROv2 arm rests on **four treated models, not six**: under our search budget the optimizer returned the baseline unchanged for two models, which we exclude rather than score as nulls (§11), and a larger budget is the obvious follow-up. Our mechanism is also demonstrably incomplete—it accounts for the DTM case but for almost none of the Turkish one (§10), where the harm recurs through a content-side route we do not identify. **Greedy decoding is not reproducible for every model on this stack**: across sessions at temperature 0 all four qwen models reproduce their CoT run item-for-item (251/251) while both gemma models do not (gemma4:e4b 90%, gemma4:31b 96%). Per-model failure counts for the gemma models therefore carry a few items of run-to-run variance—small against the effects we report ($\sim$3 items against gaps of 40+) but real, and it extends §4's instability to a stricter setting than we had claimed: for some models the harm is not stable across sessions on the *same* machine. Observational per-model cells are $n\approx100$ (65 on TurkishMMLU) and, as we show, unstable across stacks and precisions—our strong claims rest on the causal budget manipulation and the powered within-subject test, while the suggestive contrasts (reasoning-demo depression, p=.063 uncorrected) are exactly that. Pooled McNemar tests treat model$\times$item pairs as independent although items repeat across models. We therefore re-estimate our central contrast with a bootstrap that resamples items as whole clusters (Table 8), which leaves the original-stack and Turkish differentials significant and the replication-stack one marginal, and we report per-model directions and sign consistency alongside every pooled number. The differential contrast itself is post-hoc, not pre-registered. The demonstration-length manipulation could not reach the extreme-payload corner with models' own traces, and its long-native cell was under-supplied for two models—itself a finding, but a limit on the 2$\times$2's power. bf16 coverage of the small models comes from our earlier study. The large-model bf16 runs are single-seed.

Ethics Statement

DTM derives from publicly available national university-entrance examination materials and contains no personal data. Our aim is protective and self-correcting: we report that a widely deployed optimization method can manufacture a native-language harm in deployment, that two of our own earlier claims required correction (the headline's format component and the single-model full-precision erosion), and that a low-cost repair exists. All corrections are stated in the text alongside the original claims.

Reproducibility Statement

We release the instrumented harness (raw completions, token usage, finish reasons, rescue parsing), all per-item logs ($\sim$25,000 generations across six experiments), per-experiment decision memos, the frozen replication set loader, the exact-statistics module (unit-tested), and a machine-readable export of every in-text number. A verification gate asserts split fidelity with the original study's traces. Splits, option shuffling, and demonstration selection are deterministic under fixed seeds. Code and logs: `https://github.com/idrock-ai/native-language-erosion`.

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